$\pi_4(S^3) = \mathbb{Z}/2$ BY A POSTNIKOV UNIVERSAL PROPERTY: THE MAIN IDEAS OF A FORMAL PROOF

Attribution. The strategy of the proof presented here is due to David Wärn, with contributions of Owen Milner: one identifies a candidate Postnikov stage by a *marked universal property* and compares it with the truncated sphere on homotopy classes. The delooping machinery on which every step rests (Eilenberg–MacLane spaces as iterated torsors, and the criterion for a map of loop spaces to be the looping of a map [1]), and the construction of the cup product used to build the k-invariants, are also due to David Wärn. Our work was to develop this argument in detail and to formalise it in Unit; the formal proof is the development `sag/pi3-s2-postnikov`, and the detailed notes are listed in §7.

The result. In univalent type theory with replacement, and assuming only the universal property of S^3 (Axiom 1),

$$\pi_4(S^3) \cong \mathbb{Z}/2 \quad \text{as groups}$$

(`Pi4Sphere.Pi4S3.pi4-S3`, Theorem 8). No higher inductive type, no pushout, no smash product and no spectral sequence is used; the two homotopy-theoretic inputs are the cohomology computations (H3), (H4) of $K(\mathbb{Z}, 2)$, proved here from the circle and the cup square.

1. Setting

The foundations are those of `sag/em`: univalence and Rijke's replacement (images and their surjectivity) as parameters; function extensionality, propositional truncation, set quotients and the finite truncations $\|X\|_m$ are derived. Eilenberg–MacLane spaces $K(A, n)$ of an abelian group A are built as in [1]: $K(A, 1)$ is the type of A-torsors and $K(A, n{+}1)$ is the type of $K(A, n)$-torsors, i.e. of merely inhabited types with a homogeneous structure. We write $e_n : \Omega K(A, n{+}1) \simeq_* K(A, n)$, $K(h, n)$ for the map induced by a homomorphism h, and $F; G$ for composition of pointed maps (first F).

For pointed X, Y, $[X, Y]_*$ is the *set* of pointed homotopy classes, the set truncation of $\mathrm{Map}_*(X, Y)$; classes compose and form a category. Two classes are equal when their representatives are merely equal; no representative is ever chosen. Homotopy groups are $\pi_3 X = \pi_1(\Omega^2 X)$ and, by definition,

$$\pi_4 X = \pi_3(\Omega X) = \pi_1(\Omega^3 X),$$

functorial on maps and on classes (looping descends to classes since $[\Omega X, \Omega Y]_*$ is a set).

Axiom 1 (S^3)**.** A type S^3 with a point base and $\mathsf{surf} : \Omega^3(S^3, \mathsf{base})$ such that for every X, $f \mapsto (f\,\mathsf{base}, \mathsf{ap}^3_f\,\mathsf{surf})$ is an equivalence $(S^3 \to X) \simeq (x : X) \times \Omega^3(X, x)$.

It is stated as the parameters of a module, not as a global assumption. Its pointed form is $\mathrm{Map}_*(S^3, X) \simeq \Omega^3 X$, $F \mapsto (\Omega^3 F)(\mathsf{surf})$, natural in X.

2. The strategy: a marked universal property

The proof is a Yoneda argument on classes; the same argument, one level down, gives $\pi_3(S^2) = \mathbb{Z}$ (formalised first, `doc/pi3-s2-postnikov.pdf`).

Let $T = \|S^3\|_4$ with unit $\eta : S^3 \to_* T$. From the axiom, S^3 is 2-connected (into a 2-type Y evaluation at the base point is an equivalence, so $\|S^3\|_2$ is contractible), hence T is a 2-connected

Date: 28 October 2026.

4-type, η^* is a bijection on maps into 4-types, and $\pi_4\eta$ is an isomorphism (the fibres of $\Omega^k\eta$ are $(4-k)$-connected).

Proposition 2 (two instances suffice)**.** *Let P be a 4-type and $s \in [S^3, P]_*$ such that $s^* : [P, Y]_* \to [S^3, Y]_*$ is a bijection for $Y = P$ and for $Y = T$. Then $\pi_4 S^3 \cong \pi_4 T \cong \pi_4 P$.*

Proof. Let $u \in [T, P]_*$ with $[\eta]; u = s$ (η^* at P) and $t \in [P, T]_*$ with $s; t = [\eta]$ (s^* at T). Then $s; (t; u) = [\eta]; u = s = s; \mathrm{id}$, so $t; u = \mathrm{id}_P$; and $[\eta]; (u; t) = s; t = [\eta]$, so $u; t = \mathrm{id}_T$. Hence $\pi_4 u$ and $\pi_4 t$ are inverse isomorphisms. □

So the whole proof consists in constructing a 4-type P with $\pi_4 P = \mathbb{Z}/2$ and a class s with this universal property. The candidate is the delooping BR of the Postnikov stage R of $\|S^2\|_3$ with coefficients reduced mod 2.

3. The candidate BR

The cup square. Following Wärn's construction, the circle property of $K(\mathbb{Z}, 1)$ and delooping give a bipointed map $c_2 : K(\mathbb{Z}, 2) \to_* \mathrm{Map}_*(K(\mathbb{Z}, 2), K(\mathbb{Z}, 4))$, normalised so that looping it and evaluating at the generator of $\pi_1 K(\mathbb{Z}, 1)$ gives back the identity of $K(\mathbb{Z}, 2)$; the cup square is its diagonal, $\kappa(x) = c_2(x)(x)$ (`doc/cup-square.pdf`).

Cohomology of $K(\mathbb{Z}, 2)$. For every abelian group B:

- (H3) $[K(\mathbb{Z}, 2), K(B, 3)]_*$ is a point (from the circle property; `doc/circle-torsors.pdf`);
- (H4) $\Phi_B : \mathrm{Hom}(\mathbb{Z}, B) \to [K(\mathbb{Z}, 2), K(B, 4)]_*$, $h \mapsto [\kappa; K(h, 4)]$, is a bijection (`doc/h4.pdf`): every class is the diagonal of a unique bipointed map, and bipointed maps are classified by $g \mapsto c_2; K(g, 4)_*$.

With $P_2 = \mathrm{Fib}\kappa$ this gives the mapping theorem and $\pi_3(S^2) = \mathbb{Z}$.

Reduction mod 2. Let $\rho : \mathbb{Z} \to \mathbb{Z}/2$, $\bar\kappa = \kappa; K(\rho, 4) : K(\mathbb{Z}, 2) \to_* K(\mathbb{Z}/2, 4)$ and $R = \mathrm{Fib}\bar\kappa$, a simply connected 3-type with $\pi_2 R = \mathbb{Z}$ and $\pi_3 R \cong \mathbb{Z}/2$ as groups.

Theorem 3 (R deloops; `doc/cross-term.pdf`)**.** *There is $\hat F : K(\mathbb{Z}, 3) \to_* K(\mathbb{Z}/2, 5)$ with $\Omega\hat F = e_2; \bar\kappa; e_4^{-1}$. Hence $BR = \mathrm{Fib}\hat F$ is a pointed 2-connected 4-type with $\Omega BR \simeq_* R$.*

Idea. By Wärn's delooping criterion ([1], Lemma 9), $f = e_2; \bar\kappa; e_4^{-1}$ is a looping as soon as, for every loop p, f agrees with its translate $q \mapsto (fp)^{-1} \cdot f(p \cdot q)$; at these levels the space of such data is a proposition, so no coherence for the pointing in p is needed. The defect is the *cross term* "$\bar\kappa(x + y) - \bar\kappa(x) - \bar\kappa(y)$", a bipointed map; by the classification of (H4) it vanishes as soon as its diagonal "$\bar\kappa(2x) - 2\bar\kappa(x)$" does, and both terms of the diagonal vanish by construction because $2 = 0$ in $\mathbb{Z}/2$: doubling on $\Omega K(\mathbb{Z}/2, m{+}1)$ is $K(2, m) = K(0, m) = 0$, and precomposing c_2 with $K(2, 2)$ doubles its invariant. □

4. The mapping theorem for BR

A *level-3 triple* is (A, B, K) with $K : K(A, 3) \to_* K(B, 5)$; its fibre $\mathrm{Fib}K$ is a pointed 2-connected 4-type, with projection $q : \mathrm{Fib}K \to_* K(A, 3)$.

Theorem 4 (lifting)**.** *For a pointed Q, suppose (E) every $F : Q \to_* K(A, 3)$ has merely $F; K = 0$, and (U) every $d : Q \to_* \Omega K(B, 5)$ is merely 0. Then $q_* : [Q, \mathrm{Fib}K]_* \to [Q, K(A, 3)]_*$ is a bijection.*

This is the lifting lemma of the mapping theorem for $\mathrm{Fib}\kappa$ (`doc/mapping-fib.pdf`) one level up. It is applied to the two sources $Q = S^3$ and $Q = BR$. For S^3 both conditions are immediate (pointed maps out of S^3 are 3-loops, and $\Omega^3 K(B, 5)$, $\Omega^4 K(B, 5)$ are connected).

(U) for BR. Looping is an equivalence $\mathrm{Map}_*(BR, \Omega K(B,5)) \simeq \mathrm{Map}_*(R, \Omega^2 K(B,5))$, and $[R, K(B,3)]_* = 0$: a map out of $R = \mathrm{Fib}\bar{\kappa}$ restricts on the fibre to a map that deloops to some $K(g,4)$, the fibre sequence forces $[\bar{\kappa}; K(g,4)] = 0$, and

$$\Psi_B : \mathrm{Hom}(\mathbb{Z}/2, B) \to [K(\mathbb{Z},2), K(B,4)]_*, \qquad g \mapsto [\bar{\kappa}; K(g,4)] = \Phi_B(\rho; g)$$

is injective by (H4); then (H3) kills what remains (`doc/fibre-h3.pdf`, with a fibre coefficient group).

(E) for BR: the decisive point. A map $F : BR \to_* K(A,3)$ factors merely through $p : BR \to_* K(\mathbb{Z},3)$ as $p; M$ with $M : K(\mathbb{Z},3) \to_* K(B,5)$, and the obstruction is the class of $H = \mathrm{unconj}(\Omega M) : K(\mathbb{Z},2) \to_* K(B,4)$, i.e. $h \in \mathrm{Hom}(\mathbb{Z}, B)$ with $\Phi_B(h) = [H]$. The fibre sequence of BR kills it only if h factors through ρ, i.e. if $2h = 0$. So one must show that *for every target the obstruction is 2-torsion.* The point is that H is the looping of a map *by construction*:

Lemma 5 (necessity)**.** *If $\Phi_B(h) = [\mathrm{unconj}(\Omega\hat{G})]$ for some $\hat{G} : K(\mathbb{Z},3) \to_* K(B,5)$, then $2h = 0$.*

Idea. A looping commutes with squaring of loops, and squaring is $K(2,2)$ on the source and $K(2_B,4)$ on the target; so $[K(2,2)]; [H] = [H]; [K(2_B,4)]$. A computation with the bipointed classification gives $(2x)^2 = 4x^2$, i.e. $[K(2,2)]; [\kappa] = [\kappa]; [K(4,4)]$; its only new ingredient is the naturality of the invariant of a bipointed map under precomposition in the second variable. Hence $\Phi_B(4h) = \Phi_B(2h)$, and $2h = 0$ since Φ_B is injective. □

Then $h = \rho; g$, so merely $H = \bar{\kappa}; K(g,4)$, $\Omega M = \Omega(\hat{F}; K(g,5))$, and since looping is injective on maps $K(\mathbb{Z},3) \to_* K(B,5)$, merely $M = \hat{F}; K(g,5)$ and $p; M = (p; \hat{F}); K(g,5) = 0$. This is why the target is presented one level up, with a k-invariant $K(A,3) \to K(B,5)$: every obstruction is then a looping, and no coherence cell has to be normalised.

The marked class. From the axiom and the circle property one builds $\sigma_3 : S^3 \to_* K(\mathbb{Z},3)$ with $(\Omega^3\sigma_3)(\mathsf{surf})$ the generator; σ_3^* is an equivalence on maps into 3-types (a composite of five equivalences). Let

$$s = p_*^{-1}[\sigma_3] \in [S^3, BR]_*$$

(Theorem 4 at $Q = S^3$ for the triple $(\mathbb{Z}, \mathbb{Z}/2, \hat{F})$). Then s^* is a bijection into every $K(A,3)$, and the commuting square of q_* for the two sources gives:

Theorem 6 (mapping theorem)**.** *For every level-3 triple, $s^* : [BR, \mathrm{Fib}K]_* \to [S^3, \mathrm{Fib}K]_*$ is a bijection; hence so is $s^* : [BR, Y]_* \to [S^3, Y]_*$ for every Y merely pointed equivalent to some* Fib$K$*.*

5. Presentations of 2-connected 4-types

Theorem 7. *Every pointed 2-connected 4-type Y is merely pointed equivalent to* Fib$K$ *for a level-3 triple (A, B, K).*

Idea. As the presentation of $\|S^2\|_3$ (`doc/postnikov-presentation.pdf`), one level up. Let $V = \|Y\|_3$; it is $K(A,3)$ with $A = \Omega^3 V$ (a 2-connected 3-type is recognised by delooping its loop space twice). The fibres E_z of $Y \to V$ are 3-connected 4-types, hence $K(B,4)$ with $B = \Omega^3(\Omega E_{v_0})$. The structures $K(B,4) \simeq_* E_z$ form a set over Y, and a section through the given one exists since Y is 2-connected; so every E_z carries a homogeneous structure, i.e. is a $K(B,4)$-torsor, and $z \mapsto E_z$ is a map $k' : V \to K(B,5)$ with $\mathrm{Fib}k' \simeq Y$. Take $K = e_V; k'$; the pointing is only mere since Y is connected. □

The group B is not identified: for $Y = T$ it is $\pi_4 T$, the unknown. Since the goal "s^* is a bijection" is a proposition, the mere presentation of T suffices.

6. Conclusion

Theorem 8. *Under Axiom 1,* $\pi_4(S^3) \cong \mathbb{Z}/2$ *as groups.*

Proof. Theorem 6 applies to $Y = BR = \mathrm{Fib}\hat{F}$ directly and to $Y = T$ through Theorem 7. Proposition 2 with $P = BR$ gives

$$\pi_4 S^3 \cong \pi_4 T \cong \pi_4 BR = \pi_3(\Omega BR) \cong \pi_3 R \cong \mathbb{Z}/2. \qquad \square$$

7. The formalisation

The proof is checked in Unit, in `sag/pi3-s2-postnikov` (`./segc sag/pi3-s2-postnikov all`: 83 modules; a fresh check, 4038 declarations, 0 failed, about 2 s). The final declaration is

```
Pi4Sphere.Pi4S3[S3, base, surf, ump3].pi4-S3 :  GIso[pi4 S3 base, ab-grp Z2ab]
```

whose only parameters beyond the foundations (univalence, images and their surjectivity) are the S^3 axiom; the 4-truncation is constructed inside. The S^2 axiom is not used.

`CupSquare`, `H3Proof`, `H4Proof`	the cup square, (H3), (H4)
`CrossTerm`, `DeloopR`	the cross term vanishes; $\hat{F}$, BR, $\Omega BR \simeq R$
`Necessity`, `FibreH3`	Lemma 5, $(2x)^2 = 4x^2$; $[R, K(B,3)]_* = 0$
`MapBR`	Theorem 4; (E) and (U) for BR
`Sphere3Axiom`, `SourceS3`	the axiom; σ_3; s; Theorem 6
`Sphere3Trunc`	S^3 is 2-connected; $T = \|S^3\|_4$; π_4 on classes
`Presentation4`	Theorem 7; the mapping theorem at T
`Pi4Sphere`	Proposition 2; Theorem 8

Each module was written from an informal proof first, and every declaration is audited: the checker reports which definitions it had to unfold, and this is compared with a specification listing the steps the text proves "by definition" (the `.opens` files of the development, all satisfied). Detailed notes: `br-universal.pdf` (this proof in full), `cross-term.pdf`, `cup-square.pdf`, `h4.pdf`, `fibre-h3.pdf`, `circle-torsors.pdf`, `mapping-fib.pdf`, `postnikov-presentation.pdf`, `finite-truncation.pd` and `pi3-s2-postnikov.pdf` for $\pi_3(S^2) = \mathbb{Z}$.

References

[1] D. Wärn. *Eilenberg–MacLane spaces and stabilisation in homotopy type theory.* arXiv:2301.03685, 2023.